do280-4.18: Days remaining **89** | Lab hours used **4/80**

DO280-4.22: Days remaining **89** | Lab hours used **0/80**

Red Hat OpenShift Administration II: Configuring a Production Cluster

Course | Lab Environment

Chapter 10. Comprehensive Review

Summary of Learning Objectives (/rol/app/courses/do280-4.22/pages/ch10)
Lab: Set up Cluster Self-service (/rol/app/courses/do280-4.22/pages/ch10s02)
Lab: Secure Applications (/rol/app/courses/do280-4.22/pages/ch10s03)
Lab: Deploy Packaged Applications (/rol/app/courses/do280-4.22/pages/ch10s04)
Summary (/rol/app/courses/do280-4.22/pages/ch10s05)

Abstract

Goal	This lesson provides support for students who wish to prepare for the Red Hat Certified System Administrator in OpenShift (EX280) exam.
Objectives	• Validate the knowledge acquired by the student in the Red Hat OpenSh Administration II: Configuring a Production Cluster (DO280) course.

Sections	• Summary of Learning Objectives • Set up Cluster Self-service (Lab) • Secure Applications (Lab) • Deploy Packaged Applications (Lab)

Summary of Learning Objectives

Objectives

- After completing this section, you should have reviewed and refreshed the knowledge and skills that you learned in Red Hat OpenShift Administration II: Configuring a Production Cluster.

Reviewing Red Hat OpenShift Administration II: Configuring a Production Cluster (DO280)

Before beginning the comprehensive review for the *Red Hat OpenShift Administration II: Configuring a Production Cluster (DO280)* course, you should be comfortable with the topics covered in each chapter. Do not hesitate to ask the instructor for extra guidance or clarification on these topics.

Chapter 1, Declarative Resource Management

Deploy and update applications from resource manifests that are parameterized for different target environments.

- Deploy and update applications from resource manifests that are stored as YAML files.
- Deploy and update applications from resource manifests that are augmented by Kustomize.

Chapter 2, Deploying Packaged Applications

Deploy and update applications from resource manifests that are packaged for sharing and distribution.

- Deploy and update applications from resource manifests that are packaged as OpenShift templates.
- Deploy and update applications from resource manifests that are packaged as Helm charts.

Chapter 3, Authentication and Authorization

Configure authentication with the HTPasswd identity provider and assign roles to users and groups.

- Explain identity providers and configure the HTPasswd identity provider for OpenShift authentication.

- Define role-based access controls and apply permissions to users.

Chapter 4, Network Security

Protect network traffic between applications inside and outside the cluster.

- Allow and protect network connections to applications inside an OpenShift cluster.
- Restrict network traffic between projects and pods.
- Configure and use automatic service certificates.

Chapter 5, Exposing Non-HTTP/SNI Applications

Expose applications to external access without using an ingress controller.

- Expose applications to external access by using load balancer services.
- Configure user-defined networks for workload isolation and configure network attachment definitions for external access.

Chapter 6, Enabling Developer Self-service

Configure clusters for safe self-service by developers from multiple teams, and disallow self-service if operations staff must provision projects.

- Configure compute resource quotas and Kubernetes resource count quotas per project and cluster-wide.
- Configure default and maximum compute resource requirements for pods per project.
- Configure default quotas, limit ranges, role bindings, and other restrictions for new projects, and manage permissions for users to self-provision new projects.

Chapter 7, Managing Kubernetes Operators

Install and update operators that the operator lifecycle manager and the Cluster Version Operator manage.

- Describe the operator pattern and different approaches for installing and updating Kubernetes operators.
- Install and update operators by using the web console and the operator lifecycle manager.
- Install and update operators by using the operator lifecycle manager APIs.

Chapter 8, Application Security

Run applications that require elevated or special privileges from the host operating system or Kubernetes.

- Create service accounts and apply permissions, and manage security context constraints.

- Allow an application to access the Kubernetes APIs of the application's cluster.

- Automate regular cluster and application management tasks with Kubernetes cron jobs.

Chapter 9, OpenShift Updates

Update an OpenShift cluster and minimize disruption to deployed applications.

- Describe the cluster update process.
- Identify applications that use deprecated Kubernetes APIs.
- Update OLM-managed operators by using the web console and the CLI.

Revision: do280-4.22-ed1-20260827-78c07d2



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